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Bond Price Fragility and the Structure of the Mutual Fund Industry

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1 Big picture

- **Question.** When bond mutual funds face redemptions, do they sell all bonds pro rata, or do they strategically avoid selling bonds in which they own a large share of the outstanding issue?
- **Core idea.** A fund that owns a large share of a bond issue internalizes that its own sale depresses the price of the bond and hurts the value of the remaining position it continues to hold.
- **Main prediction.** During fire-sale periods, funds sell less of the bonds in which their positions are large relative to the bond's outstanding issue.
- **Data.** The paper combines Morningstar mutual fund holdings, Morningstar Direct and CRSP fund characteristics, Mergent FISD bond characteristics, and TRACE bond transactions.
- **Main sample.** January 2003–December 2019; 1,016 taxable open-end bond mutual funds; 32,006 corporate bond CUSIPs; 42,251 fund-periods; 11.3 million fund-bond-periods; 1.5 million bond-months.
- **Key variable.** *Top decile of fund position size* equals one when a fund's position in a bond is in the top decile of position size relative to bond issue size.
- **Main trading result.** During aggregate fire-sale periods, top-decile positions are sold less. The baseline coefficient is about 0.0146, meaning funds reduce these positions by about 1.46 percentage points less than other positions.
- **Mechanism.** The effect is stronger when funds have less cash/government-bond liquidity, when the position is less liquid within the fund, and when other funds in the same family also hold large positions.
- **Identification enhancements.** The paper uses fund mergers as exogenous variation in large-position ownership and the Fed's Secondary Market Corporate Credit Facility (SMCCF) as a shock that reduced expected fire-sale spillovers for eligible bonds.
- **Bond-price result.** Ownership concentration reduces bond exposure to flow-induced fire sales, especially when bonds are not protected by the Fed's intervention.
- **Fund-level result.** Funds with many large positions perform better in turmoil and have less flow sensitivity to negative performance.
- **Spillover result.** Selling pressure from low-LARGEFRAC funds hurts peer funds' returns and flows more than selling pressure from high-LARGEFRAC funds.
- **Economic takeaway.** Large ownership positions can create a stabilizing force inside the mutual fund industry. The same concentration that can look fragile from a liquidity perspective can reduce fire-sale externalities because large owners internalize price impact.

2 Data and measurement

2.1 Fund sample

- The fund sample includes open-end taxable bond mutual funds.
- All share classes with the same master portfolio are treated as one fund.
- The main sample requires funds to hold at least 50% of TNA in corporate bonds.
- Fund characteristics include TNA, family TNA, age, institutional share-class fraction, loads, flows, returns, alpha, cash share, government-bond share, corporate-bond share, and portfolio-average bond characteristics.

Interpretation. The sample focuses on bond funds that are naturally exposed to corporate-bond liquidity problems and investor redemptions.

2.2 Position-level sample

- A position is a fund-bond-period observation.
- The key position-size variable is the fund’s position divided by the bond issue size.
- The main dependent variable is position change relative to the beginning position:

$$\Delta Position_{fbt} = \frac{Position_{fbt} - Position_{fb,t-1}}{Position_{fb,t-1}}.$$

- The indicator *Top decile of fund position size* equals one for large positions in the fund-bond-time distribution.

Interpretation. A large position is not necessarily large for the fund, but large relative to the bond issue. That is the dimension that matters for price impact.

2.3 Bond-level sample

- Bonds are nonputtable U.S. corporate debentures and corporate bank notes held by at least one sample fund.
- Bond characteristics include maturity, issue size, age, investment-grade status, upgrades/downgrades, mutual fund ownership, top fund ownership, top family ownership, insurance ownership, turnover, zero-trading days, effective spreads, and returns.
- Bond returns are calculated from TRACE transaction prices.

Interpretation. Bond-level ownership measures let the paper ask whether concentrated mutual-fund ownership affects the transmission of fire-sale shocks into bond prices.

2.4 Flow-induced fire sales and turmoil

- Periods of financial turmoil are months in which aggregate monthly flows to taxable bond mutual funds are in the bottom decile.
- For individual-flow tests, the paper also examines fund-periods in which individual fund flows are in the bottom quintile.
- Flow-induced fire sales (FIFS) proxy for selling pressure from fund outflows and are used to test price spillovers.

Interpretation. The paper separates the trading decision from the price effect. First it asks how funds sell; then it asks whether the ownership structure of selling funds affects bond returns and peer-fund outcomes.

2.5 COVID and SMCCF sample

- The COVID-19 sample runs from April 2020 to September 2020.
- The Fed’s SMCCF made some investment-grade bonds eligible for a price backstop.
- The paper compares Fed-eligible bonds with ineligible investment-grade and high-yield bonds.

Interpretation. The SMCCF test is powerful because it changes the expected downside from fire-sale spillovers. If large owners sell less because they internalize price impact, the effect should weaken for Fed-protected bonds.

3 Models and empirical specifications

3.1 Conceptual framework: fire-sale price without concentration

For a representative security with atomistic owners, if a fraction ρ of investors must sell, the fire-sale price is

$$P = 1 - h\rho, \quad (1)$$

where $h > 0$ captures price impact.

Interpretation. Atomistic investors take the price impact as given. No single investor internalizes the price effect of its own trade.

3.2 Security with concentrated ownership

For a security in which one fund owns share ξ , if atomistic owners sell $(1 - \xi)\rho$ and the large owner sells δ_1 , the price is

$$P^* = 1 - h[(1 - \xi)\rho + \delta_1]. \quad (2)$$

The `HIGHLARGEFRAC` fund chooses sales of the concentrated security, δ_1 , and sales of other securities, δ_2 , to maximize

$$[1 - h((1 - \xi)\rho + \delta_1)](\xi - \delta_1) + (1 - h\rho)(X - \delta_2), \quad (3)$$

subject to the redemption constraint

$$[1 - h((1 - \xi)\rho + \delta_1)]\delta_1 + (1 - h\rho)\delta_2 = R. \quad (4)$$

Interpretation. The large owner suffers twice from selling the concentrated bond: it receives a lower price on the amount sold, and it marks down the remaining position it still owns.

3.3 Proposition 1

Proposition. During fire sales, holding constant the redemptions experienced by the owners of different securities, a fund sells a lower proportion of the security in which it holds a large share of the outstanding issue than of securities it holds in small amounts.

Interpretation. The theory predicts stabilizing trading by large owners. They do not sell less because they are passive; they sell less because price impact is costly to them.

3.4 Main trading regression

The main position-level specification is

$$PositionChange_{fbt} = \alpha + \beta TopDecilePositionSize_{fbt} + \mathbf{1}_{bt} + \gamma_{ft} + \varepsilon_{fbt}. \quad (5)$$

- $\mathbf{1}_{bt}$ are bond-by-time fixed effects.
- γ_{ft} are fund-by-time fixed effects.
- The sample is restricted to turmoil periods.
- $\beta = 0$ means pro-rata selling.
- $\beta > 0$ means funds sell large issue-share positions less.

Interpretation. Fund-by-time fixed effects absorb the fund's redemption shock and overall need to raise cash. Bond-by-time fixed effects absorb bond-specific conditions. Identification comes from within-fund, within-bond-time variation across positions.

3.5 Outflow sensitivity regression

To connect trading to fund redemptions, the paper estimates regressions with outflows and their interaction with large position size:

$$PositionChange_{f_{bt}} = \alpha + \beta_1 Outflow_{f_t} + \beta_2 (Outflow_{f_t} \times TopDecilePositionSize_{f_{bt}}) + Controls + FE + \varepsilon_{f_{bt}}. \quad (6)$$

Interpretation. $\beta_1 < 0$ captures forced selling from outflows. A positive β_2 means that even when the fund is forced to sell, it sells large issue-share positions less.

3.6 Bond return regressions

At the bond-month level, the paper studies whether concentrated ownership weakens the effect of selling pressure on returns:

$$Return_{bt} = \alpha + \beta_1 FIFS_{bt} + \beta_2 TopFundOwnership_{bt} + \beta_3 (FIFS_{bt} \times TopFundOwnership_{bt}) + Controls + FE + \varepsilon_{bt}. \quad (7)$$

Interpretation. If large owners internalize fire-sale spillovers, then $\beta_3 > 0$: flow-induced selling pressure should be less damaging when ownership is concentrated.

3.7 Fund performance regression

The fund-level performance equation is

$$Alpha_{ft} = \alpha + \beta LARGEFRAC_{f,t-1} + \gamma' Controls_{f,t-1} + FE + \varepsilon_{ft}, \quad (8)$$

where *LARGEFRAC* is the fraction of the fund's portfolio in top-decile positions.

Interpretation. Funds with more large issue-share positions should perform better in turmoil because they can reduce the price impact of their own sales.

3.8 Flow sensitivity regression

To test financial fragility, the paper estimates

$$Flow_{ft} = \alpha + \beta_1 Alpha_{f,t-1} + \beta_2 LARGEFRAC_{f,t-1} + \beta_3 (Alpha_{f,t-1} \times LARGEFRAC_{f,t-1}) + Controls + FE + \varepsilon_{ft}. \quad (9)$$

Interpretation. For funds with negative alpha, a negative interaction means that flows are less sensitive to poor performance when the fund has more large positions.

4 Identification and empirical design

4.1 Main within-fund design

- The trading regressions compare different positions within the same fund during the same period.
- Fund-by-period fixed effects absorb the fund's redemption shock, cash need, portfolio policy, and management style in that period.
- Bond-by-period fixed effects absorb bond liquidity, credit risk, return, rating news, and market conditions in that period.

Interpretation. The coefficient asks: among bonds held by the same fund at the same time, and comparing within the same bond-time environment, does the fund sell less of bonds in which it owns a large share?

4.2 Mechanism tests

- If funds are selling less to avoid price impact, the effect should be stronger when other liquidity sources are scarce.
- The paper interacts large-position indicators with low cash/government-bond holdings, within-fund bond illiquidity, and other same-family holdings.
- All mechanisms point toward internalization of price spillovers rather than simple mechanical portfolio constraints.

Interpretation. The theory is about private incentives to limit price impact. The mechanism tests show the result becomes stronger when price impact matters more.

4.3 Fund merger IV

- Fund mergers mechanically change a fund’s position size relative to bond issue size.
- The paper uses post-merger combined positions as an instrument for top-decile position size.
- The first-stage F-statistic is strong in the reported test.

Interpretation. The merger IV addresses the concern that funds deliberately choose large positions in bonds they would have sold less anyway.

4.4 SMCCF test

- The Fed’s SMCCF reduced downside price pressure for eligible bonds.
- If large-owner behavior is driven by fear of fire-sale spillovers, it should be weaker for Fed-eligible bonds.
- The paper finds that the stabilizing trading effect is weak or reversed for eligible bonds and strong for ineligible bonds.

Interpretation. This is a key mechanism validation: when the negative externality is reduced by a policy backstop, the incentive to avoid selling large positions also weakens.

5 Tables and figures

The paper contains empirical tables rather than plotted figures. The tables below are directly cropped from the original paper and grouped compactly.

5.1 Table 1: Summary statistics

Read:

- Panel A reports 42,251 fund-periods in the main sample. Mean fund TNA is \$1.246 billion, and the mean fund holds 77.5% of its portfolio in corporate bonds.
- The average value of LARGEFRAC is 0.174; its 90th percentile is 0.665.
- Panel B reports 11.3 million fund-bond-periods. The mean position size is 0.636% of the bond issue size.
- Panel C reports 1.5 million bond-months. Mean mutual fund ownership is 10%; mean top-fund ownership is 3.5%.

Economic message. The setting has enough cross-sectional variation in fund-level concentrated positions to study whether large owners behave differently during redemptions.

Table 1
Summary statistics
A. Fund-level variables

Variable	Main sample (42,251 fund-periods)					COVID sample (4,785 fund-periods)
	Mean	Standard Deviation	PCT 10	PCT 50	PCT 90	Mean
Total net assets (TNA) (\$ mil.)	1,246	2,681	26	334	3,103	2,709
Family TNA (\$ mil.)	26,150	61,522	176	8,400	57,801	62,016
Fund age (year)	12.330	9.992	1.837	9.706	27.088	16.386
Institutional share class fraction	0.506	0.394	0.000	0.508	1.000	0.674
Avg. maximum rear load ($\times 100$)	0.375	0.638	0.000	0.013	1.747	0.071
Flow ($\times 100$)	0.603	3.362	-3.017	0.071	5.207	0.663
Return ($\times 100$)	0.476	1.268	-1.034	0.473	1.979	1.359
Alpha (%)	-0.053	0.807	-0.599	-0.019	0.529	0.065
Broker affiliation	0.074	0.262	0.000	0.000	0.000	0.044
Cash as % of portfolio	6.640	9.641	0.629	4.996	15.369	9.212
Government bonds as % of portfolio	5.601	11.298	0.000	0.000	19.347	18.816
Corporate bonds as % of portfolio	77.485	18.465	50.978	84.255	95.370	45.642
Portfolio avg. bond turnover	0.099	0.028	0.061	0.103	0.132	0.071
Portfolio avg. fraction of zero trading days	0.458	0.129	0.323	0.438	0.643	0.481
Portfolio avg. effective spread ($\times 100$)	0.507	0.237	0.281	0.454	0.791	0.533
Portfolio avg. bond issue size (\$ mil.)	872	307	484	855	1,265	1,184
Portfolio avg. bond age (year)	2.818	1.284	1.516	2.563	4.490	3.489
Portfolio avg. coupon rate	6.568	1.925	3.966	6.569	8.910	4.587
Portfolio avg. credit rating (1 = AAA)	12.877	3.787	7.000	15.000	16.000	9.920
Portfolio Effective duration (year)	4.321	2.043	2.344	3.985	6.566	5.890
Top dec. positions/TNA (LARGEFRAC)	0.174	0.272	0.000	0.009	0.665	0.116

B. Position-level variables

Variable	Main sample (11,312,852 fund-bond-periods)					COVID sample (973,077 fund-bond-periods)
	Mean	Standard Deviation	PCT 10	PCT 50	PCT 90	Mean
Position size/Bond issue size ($\times 100$)	0.636	1.535	0.017	0.200	1.525	0.486
Family position size/Bond issue size ($\times 100$)	1.584	3.589	0.042	0.592	4.173	1.642
Position change/Beginning position ($\times 100$)	-4.611	27.927	-5.656	0.000	0.000	-4.911
Position increase	0.063	0.243	0.000	0.000	0.000	0.070
Position decrease	0.109	0.312	0.000	0.000	1.000	0.101

(a) Table 1, panels A–B

Table 1
(Continued)
C. Bond-level variables

Variable	Main sample (1,514,632 bond-months)					COVID sample (51,597 bond-months)
	Mean	Standard Deviation	PCT 10	PCT 50	PCT 90	Mean
Bond maturity (year)	8.699	8.301	1.498	5.916	24.287	9.645
Bond issue size (\$ mil.)	593	504	200	450	1,250	767
Bond age (year)	4.466	4.144	0.550	3.274	9.621	5.191
Investment grade	0.666	0.471	0.000	1.000	1.000	0.770
Upgrade	0.021	0.142	0.000	0.000	0.000	0.029
Downgrade	0.010	0.082	0.000	0.000	0.000	0.009
Mutual fund ownership	0.100	0.097	0.007	0.068	0.244	0.107
Top fund ownership	0.035	0.037	0.004	0.022	0.080	0.034
Top family ownership	0.046	0.044	0.005	0.033	0.100	0.053
Insurance ownership	0.254	0.237	0.000	0.189	0.611	0.223
Turnover	0.062	0.060	0.003	0.043	0.160	0.040
Fraction of zero trading days	0.574	0.290	0.187	0.592	0.961	0.618
Effective (bid-ask) spread ($\times 100$)	0.694	0.693	0.111	0.452	1.654	0.799
Flow-induced fire sales (FIFS) ($\times 100$)	0.036	0.077	0.000	0.000	0.084	0.032
JLSW fragility	0.841	0.476	0.409	0.726	1.349	—
Return ($\times 100$)	0.293	2.723	-2.036	0.231	2.766	1.829
Return on matched IG bond index ($\times 100$)	0.269	1.421	-1.314	0.391	1.848	1.798
Return on matched HY bond index ($\times 100$)	0.464	2.175	-1.972	0.682	2.437	2.547

This table presents summary statistics for fund-level (panel A), position-level (panel B), and bond-level (panel C) variables. The data on fund holdings and characteristics are from Morningstar, Morningstar Direct, and CRSP. The data on bond characteristics are from Mergent FISD. The data on corporate bond transactions, which we use to calculate bond prices and returns, are from FINRA's Enhanced TRACE. The main sample covers the period from January 2003 to December 2019. The main fund sample includes only open-ended taxable bond mutual funds that hold at least 50% of the total net assets under management (TNA) in corporate bonds. All share classes that share the same master portfolio count as one fund, and the number of unique funds is 1,016. The bond sample includes only nonputtable U.S. Corporate Debentures and U.S. Corporate Bank Notes (bond type CDEB or USBN) that are held by at least one fund on the latest report date, and the number of unique bond CUSIPs is 32,006. The position sample includes only the positions of sample funds in sample bonds. The COVID-19 sample covers the period from April 2020 to September 2020, with 1,159 unique funds and 10,385 unique bond CUSIPs. The sample criteria are the same as for the main sample, except without the requirement that funds must hold at least 50% of TNA in corporate bonds. In panel A, the observations represent 12 bond

(b) Table 1, panel C

5.2 Tables 2–3: Main trading effect and mechanisms

Read:

- Table 2 shows that top-decile positions are sold less during aggregate turmoil. The all-bond coefficient is 0.0146** in column 1 and remains positive with controls and fixed effects.
- The effect is stronger for high-yield bonds than investment-grade bonds.
- Table 3 shows the mechanism: the effect is stronger when funds have less cash/government bonds, when the position is less liquid within the fund, and when other funds in the family also have large positions.

Economic message. The main effect is not just a liquidity-ranking story. Funds appear to protect positions whose sale would create large price spillovers.

Table 2
Fund trading during periods of financial turmoil and the fund's share of an outstanding bond issue

	(1) ALL	(2) IG	(3) HY	(4) ALL	(5) IG	(6) HY	(7) ALL
Top decile of fund position size	0.0146** (0.006)	0.0065 (0.006)	0.0196** (0.008)	0.0122** (0.005)	0.0054 (0.004)	0.0145** (0.006)	0.0148** (0.006)
Within-fund-period turnover decile							-0.0016** (0.001)
Bond controls							0.0149 (0.009)
Insurance ownership	0.0166* (0.009)	0.0021 (0.009)	0.0207 (0.019)				-0.0961** (0.040)
Turnover	-0.1536*** (0.032)	-0.0945*** (0.029)	-0.1687*** (0.040)				-0.0130** (0.006)
Fraction of zero trading days	-0.0115* (0.006)	0.0131** (0.005)	-0.0242*** (0.008)				0.5562** (0.242)
Effective spread	0.5745** (0.235)	1.0206*** (0.260)	0.5560 (0.368)				0.0286*** (0.005)
ln(1 + Bond maturity)	0.0286*** (0.005)	0.0285*** (0.007)	0.0317*** (0.005)				-0.0024 (0.002)
ln(1 + Bond issue size)	-0.0024 (0.002)	-0.0002 (0.002)	-0.0036 (0.003)				-0.0097*** (0.002)
ln(1 + Bond age)	-0.0097*** (0.002)	-0.0038* (0.002)	-0.0138*** (0.003)				-0.0221** (0.009)
Investment grade	-0.0221** (0.009)						-0.1598*** (0.052)
Upgrade	-0.1598*** (0.052)		-0.1611*** (0.052)				-0.2372*** (0.056)
Downgrade	-0.2372*** (0.056)	-0.2313*** (0.055)					
Fund × Period fixed effects	YES	YES	YES	YES	YES	YES	YES
Bond × Period fixed effects	NO	NO	NO	YES	YES	YES	NO
Observations	692,267	245,064	447,203	698,913	233,382	465,531	692,267
R-squared	.101	.137	.109	.360	.425	.354	.101

This table reports OLS estimates for panel regressions of fund position changes in turmoil periods on fund position size. Financial turmoil is defined as periods in which aggregate monthly flows to all taxable bond mutual funds are in the bottom decile. The dependent variable, position change, is measured as a fraction of the position on the previous report date, and hence the sample includes only existing positions. Appendix B defines all variables. All columns include fund-reporting period fixed effects. Columns 4-6 also include bond-period fixed effects. Standard errors, double-clustered by fund family and reporting period, are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

(a) Table 2

Table 3
Mechanism tests for fund trading during periods of financial turmoil

	(1)	(2)	(3)
Top decile of fund position size	0.0041 (0.004)	0.0107* (0.005)	0.0136** (0.006)
Low cash and government bonds as % of portfolio × Top decile of fund position size	0.0145** (0.007)		
High within-fund-period effective spread rank		0.0050** (0.002)	
High within-fund-period effective spread rank × Top decile of fund position size		0.0065** (0.003)	
Top decile of position size of "other" funds in the same family			0.0110*** (0.004)
Bond controls	YES	YES	YES
Fund × Period fixed effects	YES	YES	YES
Bond × Period fixed effects	NO	NO	NO
Observations	692,267	692,267	692,267
R-squared	.101	.101	.101

This table reports OLS estimates for panel regressions of fund position changes during turmoil periods on fund position size. Columns 1 and 2 also include the interaction of position size with asset liquidity, and column 3 adds position size of all other funds in the same family. Financial turmoil is defined as periods in which aggregate monthly flows to all taxable bond mutual funds are in the bottom decile. The dependent variable, position change, is measured as a fraction of the position on the previous report date, and hence the sample includes only existing positions. Low cash and government bonds as % of portfolio is a dummy that equals one if the fund's holdings of cash and government bonds are in the bottom two-thirds of the sample. High within-fund-period effective spread rank is a dummy that equals one if the bond position ranks in the top two-thirds for effective spread among all bond positions in the fund's portfolio on the previous report date. Top decile of position size of "other" funds in the same family is a dummy that equals one if the aggregate position size of all other funds (i.e., excluding the fund itself) in the fund's family ranks in the top decile of the sample. Appendix B defines all variables. All columns include fund-reporting period fixed effects. Standard errors, double-clustered by fund family and

(b) Table 3

5.3 Tables 4–5: Outflow sensitivity and merger IV

Read:

- Table 4 shows that outflows lead to selling, but the outflow effect is softened for top-decile positions. The interaction of outflows with top-decile position size is positive and significant.
- Table 5 uses fund mergers as an instrument. The first-stage coefficient on $Treated \times \ln(1 + Fund\ TNA)$ is 0.0031*** and the Kleibergen-Paap F-statistic is 30.72.
- The second-stage coefficient on top-decile position size is 0.2859*, larger than OLS, consistent with stronger effects for merger-induced large positions.

Economic message. Even when funds face severe redemption pressure, they strategically preserve large issue-share positions. The merger IV supports a causal interpretation.

Table 4 Fund's share of an outstanding bond issue and the sensitivity of fund trading to outflows								
	Stress × Periods with aggregate flows in bottom decile ($\leq -0.85\%$, average = -1.388%)				Stress × Fund-periods with individual flows in bottom quantile ($\leq -1.62\%$, average = -3.482%)			
	(1) ALL	(2) IG	(3) HY	(4) ALL	(5) ALL	(6) IG	(7) HY	(8) ALL
Outflow	-0.7964*** (0.057)	-0.7612*** (0.062)	-0.8244*** (0.060)	-0.7160*** (0.072)	-1.3784*** (0.118)	-1.8175*** (0.149)	-1.2487*** (0.144)	-1.3006*** (0.155)
Outflow × Top decile of fund position size	0.2802*** (0.057)	0.1198* (0.072)	0.2737*** (0.068)	0.2437*** (0.056)	0.1544** (0.062)	0.1792* (0.101)	0.1637** (0.070)	0.1526** (0.061)
Outflow × Within-fund-period turnover decile				-0.0148 (0.010)				-0.0129* (0.007)
Bond controls	YES	YES	YES	YES	YES	YES	YES	YES
Time-varying fund controls	YES	YES	YES	YES	YES	YES	YES	YES
Fund fixed effects	YES	YES	YES	YES	YES	YES	YES	YES
Period fixed effects	YES	YES	YES	YES	YES	YES	YES	YES
Observations	676,340	242,194	434,146	676,340	1,254,138	315,080	939,058	1,254,138
R-squared	.066	.090	.069	.066	.066	.097	.062	.066

This table reports OLS estimates for panel regressions of fund position changes in turmoil periods on fund outflows and the interaction of fund outflows with position size. In columns 1-4, the sample includes only periods of financial turmoil, defined as periods in which aggregate monthly flows to all taxable bond mutual funds are in the bottom decile. In columns 5-8, financial turmoil is measured using individual fund flows, and the sample includes only the fund-period observations for which the fund flows are in the bottom quantile of the sample. The dependent variable is the position change as a fraction of the position on the previous report date, and hence the sample includes only existing positions. Appendix B defines all variables. All columns include bond controls (as in Table 2), time-varying fund controls including portfolio time-varying characteristics (% cash, % government bonds, % corporate bonds, average bond turnover, average fraction of zero trading days, average effective spread, average coupon rate, average credit rating, effective duration, natural logarithm of one plus fund average bond issue size, and natural logarithm of one plus the average bond age), fund time-varying characteristics (age, size, family size, institutional share class fraction, and average maximum rear load), and fund and reporting period fixed effects. Standard errors, clustered by fund family, are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 5
Exogenous variation in a fund's share of an outstanding bond issue

	(1) 1st stage Top decile of fund position size	(2) 2nd stage Position change
Dependent variable:		
Top decile of fund position size		0.2859* (0.157)
$Treated \times \ln(1 + Fund\ TNA)$	0.0031*** (0.001)	
Bond controls	YES	YES
Fund × Period fixed effects	YES	YES
Kleibergen-Paap rk Wald F-statistic	30.72	
Observations	21,958	21,958

This table reports 2SLS estimates for a panel IV regression of fund position changes in turmoil periods on fund position size. Fund mergers are used as a source of exogenous variation in fund position size. The sample includes only the funds that experience a merger in the past 12 months and the periods in which the aggregate flows are in the bottom decile from January 2003 to December 2019. The dependent variable, position change, is measured as a fraction of the position on the previous report date, and hence only existing positions are considered. Top decile of fund position size is instrumented by the product of the $Treated$ dummy, which equals one if the bond position comes from both the acquirer and target portfolios, and natural logarithm of the (merged) fund's TNA in the period immediately after the merger. Appendix B defines all variables. Column 1 reports the first-stage estimates. Column 2 reports the second-stage estimates. As in Table 2, both columns include bond controls and fund-reporting period fixed effects. Standard errors, clustered by reporting period, are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.4 Table 6: Fed SMCCF and trading behavior

Read:

- Panel A shows that top-decile positions are sold less overall after the SMCCF, but the coefficient is negative for Fed-eligible bonds and positive for Fed-ineligible bonds.
- For Fed-ineligible IG bonds, the coefficient is 0.0273***; for Fed-ineligible HY bonds, it is 0.0302*.
- Panel B shows that the interaction between top-decile position size and Fed eligibility is negative and significant, largely offsetting the main top-decile effect.

Economic message. When the Fed provided a backstop, the incentive for large owners to avoid selling those bonds weakened. This supports the price-spillover mechanism.

Table 6
Fund trading in response to the Fed's SMCCF
A. Different subsamples of corporate bonds

	(1) ALL	(2) Fed Eligible	(3) Fed Ineligible IG	(4) Fed Ineligible HY
<i>Top decile of fund position size</i>	0.0178* (0.008)	-0.0140** (0.005)	0.0273*** (0.005)	0.0302* (0.012)
<i>Bond controls</i>				
<i>Insurance ownership</i>	-0.0407** (0.016)	-0.0035 (0.006)	0.0137 (0.032)	0.0188 (0.050)
<i>Turnover</i>	0.1193* (0.055)	-0.0102 (0.016)	0.1639** (0.063)	-0.0202 (0.053)
<i>Fraction of zero trading days</i>	0.0809*** (0.017)	0.0034 (0.005)	0.1204*** (0.016)	0.0080 (0.019)
<i>Effective spread</i>	0.4980 (0.647)	0.5255** (0.181)	1.8773** (0.697)	0.1151 (0.847)
<i>ln(1 + Bond maturity)</i>	0.0350*** (0.007)	-0.0031 (0.003)	0.1120*** (0.012)	0.0552*** (0.013)
<i>ln(1 + Bond issue size)</i>	-0.0081* (0.003)	-0.0034 (0.003)	0.0129 (0.007)	-0.0112* (0.005)
<i>ln(1 + Bond age)</i>	0.0085 (0.005)	-0.0034* (0.001)	-0.0018 (0.005)	-0.0174 (0.010)
<i>Investment grade</i>	-0.0197 (0.014)			
<i>Upgrade</i>	-0.0586 (0.040)			-0.0311 (0.034)
<i>Downgrade</i>	-0.1380** (0.046)	-0.0821** (0.030)	-0.1148 (0.068)	
Fund x Period fixed effects	YES	YES	YES	YES
Observations	91,381	20,674	47,109	23,598
R-squared	.121	.210	.230	.181

(Continued)

Table 6
(Continued)
B. Differences in trading in fed eligible and ineligible corporate bonds

	(1) IG	(2) IG (2–8 years)	(3) IG (4–6 years)
<i>Top decile of fund position size</i>	0.0266** (0.007)	0.0431*** (0.009)	0.0540** (0.017)
<i>Fed eligible</i>	0.2711*** (0.026)	0.2884*** (0.034)	0.3119*** (0.042)
<i>Top decile of fund position size × Fed eligible</i>	-0.0402* (0.017)	-0.0699*** (0.009)	-0.0675*** (0.015)
<i>Bond controls</i>			
<i>Insurance ownership</i>	-0.0001 (0.020)	-0.0813** (0.024)	-0.1323** (0.038)
<i>Turnover</i>	0.1493** (0.057)	0.0544 (0.033)	0.1198 (0.063)
<i>Fraction of zero trading days</i>	0.1027*** (0.014)	0.1467*** (0.020)	0.1661** (0.041)
<i>Effective spread</i>	1.7786** (0.593)	2.4367 (1.316)	2.2197 (2.899)
<i>ln(1 + Bond maturity)</i>	0.1172*** (0.013)	0.3645*** (0.055)	1.2222*** (0.196)
<i>ln(1 + Bond issue size)</i>	0.0086 (0.005)	0.0085 (0.004)	0.0080 (0.005)
<i>ln(1 + Bond age)</i>	0.0015 (0.003)	0.0197** (0.006)	0.0294** (0.009)
<i>Downgrade</i>	-0.1157* (0.052)	-0.0568 (0.057)	-0.0552 (0.088)
Fund × Period fixed effects	YES	YES	YES
Observations	66,738	24,124	9,187
R-squared	.202	.289	.314

(a) Table 6, panel A

(b) Table 6, panel B

5.5 Tables 7–10: Bond returns and ownership concentration

Read:

- Table 7 panel A shows that bonds in the highest top-fund-ownership decile have smaller fire-sale return losses than bonds in the lowest decile.
- Table 7 panel B shows that the interaction $FIFS \times Top\ fund\ ownership$ is positive in several specifications, meaning ownership concentration mitigates fire-sale price pressure.
- Table 8 uses mergers as IV variation in top fund ownership. The second-stage interaction coefficient is 35.7742**.
- Table 9 shows similar patterns during the SMCCF episode, with effects concentrated in Fed-ineligible bonds.
- Table 10 shows that concentrated ownership also helps when aggregate mutual-fund outflows are severe.

Economic message. Ownership concentration does not merely change trading. It changes prices: bonds with concentrated fund ownership are less exposed to flow-induced fire-sale losses.

Table 7
Bond returns and ownership concentration during periods of financial turmoil
A. Average bond returns in turmoil and nonturmoil periods

FIFS in top decile?	Decile of top fund ownership									
	1 (Lowest)	2	3	4	5	6	7	8	9	10 (Highest)
<i>All bonds</i>										
1	−0.927%	−0.529%	−0.470%	−0.393%	−0.389%	−0.397%	−0.392%	−0.385%	−0.364%	−0.314%
0	0.127%	0.118%	0.123%	0.132%	0.135%	0.135%	0.147%	0.152%	0.126%	0.130%
1-0										0.613%***
										0.002%
										0.611%***
<i>Investment-grade bonds</i>										
1	−0.290%	−0.104%	−0.092%	−0.081%	−0.064%	−0.066%	−0.050%	−0.043%	−0.042%	−0.027%
0	0.032%	0.050%	0.050%	0.050%	0.057%	0.049%	0.059%	0.051%	0.050%	0.052%
1-0										0.262%***
										0.020%
										0.242%***
<i>High-yield bonds</i>										
1	−1.485%	−1.062%	−0.807%	−0.729%	−0.772%	−0.702%	−0.713%	−0.714%	−0.607%	−0.537%
0	0.311%	0.308%	0.285%	0.284%	0.312%	0.318%	0.335%	0.331%	0.334%	0.320%
1-0										0.948%***
										0.009%
										0.939%***

(Continued)

Figure 4: Table 7, panel A: bond returns by ownership concentration

Table 7 (Continued)						
B. Regressions of bond returns on flow-induced fire sales and its interaction with ownership concentration						
	(1) ALL	(2) ALL	(3) IG	(4) HY	(5) HY	(6) HY
Turnover	0.0053*** (0.002)	0.0053*** (0.002)	0.0036*** (0.001)	0.0078*** (0.002)	0.0022 (0.002)	0.0078*** (0.002)
Fraction of zero trading days	0.0006 (0.000)	0.0006 (0.000)	0.0006 (0.000)	0.0022** (0.001)	−0.0013** (0.001)	0.0022** (0.001)
Effective spread	0.1541*** (0.001)	0.1543*** (0.001)	0.1291*** (0.001)	0.2046*** (0.001)	0.0550** (0.001)	0.2034*** (0.001)
ILSW fragility	0.020 (0.001)	0.020 (0.001)	0.020 (0.001)	0.020 (0.001)	0.020 (0.001)	0.020 (0.001)
ln(1 + Bond maturity)	−0.0009 (0.001)	−0.0009 (0.001)	−0.0009 (0.001)	−0.0027*** (0.001)	−0.0001 (0.001)	−0.0023** (0.001)
ln(1 + Bond issue size)	0.0001 (0.001)	0.0001 (0.001)	0.0001 (0.001)	0.0001 (0.001)	0.0001 (0.001)	0.0001 (0.001)
ln(1 + Bond age)	0.0001 (0.001)	0.0001 (0.001)	0.0001 (0.001)	0.0001 (0.001)	0.0001 (0.001)	0.0001 (0.001)
Investment grade	−0.0020** (0.001)	−0.0020** (0.001)	−0.0008*** (0.000)	−0.0008*** (0.000)	−0.0008*** (0.000)	−0.0008*** (0.000)
Upgrade	0.0008*** (0.000)	0.0008*** (0.000)	0.0008*** (0.000)	0.0012* (0.001)	0.0006 (0.001)	0.0012* (0.001)
Downgrade	−0.0074*** (0.002)	−0.0074*** (0.002)	−0.0074*** (0.002)	−0.0048** (0.002)	−0.0008 (0.002)	−0.0048** (0.002)
Issuer fixed effects	YES	YES	YES	YES	NO	YES
Month fixed effects	YES	YES	YES	YES	NO	YES
Issuer x Month fixed effects	NO	NO	NO	NO	YES	NO
Observations	911,316	911,316	640,885	276,431	189,633	276,431
R-squared	352	352	432	310	779	310

Panel A reports average bond returns in turmoil and nonturmoil periods for bond portfolios in different deciles of ownership by the top fund. The sample includes all bond-month observations, and the turmoil indicator equals 1 if flow-induced fire sales (FIFS) are in the top decile based on a pooled sorting of the sample. Tests of differences are conducted using standard errors clustered by month. Panel B reports OLS estimates for panel regressions of bond returns on FIFS, top fund ownership, and their interaction. All columns include market and bond controls, and except column 5, month and issuer fixed effects. Column 5 includes issuer-month fixed effects. Standard errors, clustered by month, are in parentheses. For each bond-month observation, return is calculated as the percentage change in the volume-weighted average price (VWAP) of the bond from the last day on which there are transactions in the previous month to the last day on which there are transactions in the current month.

(a) Table 7, panel B, first page

(b) Table 7, panel B, continued

Table 8

Exogenous variation in ownership concentration and bond returns during periods of turmoil

Dependent variable:	(1) 1st stage <i>Top fund ownership</i>	(2) 1st stage <i>FIFS × Top fund ownership</i>	(3) 2nd stage <i>Bond return</i>
<i>Flow-induced fire sales (FIFS)</i>	−3.7736*** (0.231)	0.0513*** (0.001)	−3.5188*** (1.131)
<i>Top fund ownership</i>			0.0685 (0.070)
<i>FIFS × Top fund ownership</i>			35.7742** (17.520)
<i>Merged position/Issue size</i>	0.0725*** (0.011)	−0.0002*** (0.000)	
<i>FIFS × Merged position/Issue size</i>	24.7102*** (9.360)	0.4547*** (0.023)	
Market controls	YES	YES	YES
Bond controls	YES	YES	YES
Issuer fixed effects	YES	YES	YES
Month fixed effects	YES	YES	YES
Kleibergen-Paap rk Wald F-statistic		41.49	
Observations	333,555	333,555	333,555

This table reports estimates for a panel IV regression of bond returns on FIFS as a proxy for market turmoil, top fund ownership, and their interaction. Fund mergers are used as a source of exogenous variation in top fund ownership. The sample includes only the bonds that are in the portfolios of either the acquirer or target funds immediately before each merger and only the periods from months $t+1$ to $t+12$, where month t is the merger event month. Only the mergers in which the merged funds' TNAs are at least \$600 million (25th percentile of the mergers sample) are used. For each bond-month observation, the dependent variable, bond return is calculated as the percentage change in the volume-weighted average price (VWAP) of the bond from the last day on which there are transactions in the previous month to the last day on which there are transactions in the current month. Only returns calculated from VWAP that lie in the last 10 days of each month are used. Top fund ownership is instrumented by the sum of positions in a given bond by the acquirer and target funds divided by the bond's issue size. If only the acquirer fund or the target fund holds the bond, the instrument takes the value of zero. Appendix B defines all variables. Columns 1 and 2 report the first-stage estimates, where the dependent variables are top fund ownership and the interaction of top fund ownership with FIFS. Column 3 reports the second-stage estimates. All columns include bond controls, and month and issuer fixed effects. Standard errors, clustered by month, are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

(a) Table 8

Table 9

Bond returns and ownership concentration after the introduction of the Fed's SMCCF

A. Different subsamples of corporate bonds

	(1) ALL	(2) Fed Eligible	(3) Fed Ineligible IG	(4) Fed Ineligible HY
<i>Flow induced fire sales (FIFS)</i>	−3.1905*** (0.549)	1.1611 (0.807)	−3.9907*** (0.992)	−2.9218** (1.349)
<i>Top fund ownership</i>	−0.0262*** (0.007)	0.0062 (0.005)	−0.0278** (0.011)	−0.0062 (0.020)
<i>FIFS × Top fund ownership</i>	27.3694*** (8.367)	−17.9510** (8.574)	45.0694** (17.809)	36.9122* (20.432)
Market controls				
<i>Matched index return</i>	−0.8658*** (0.050)	−0.0650 (0.065)	−1.4225*** (0.069)	−0.3171 (0.206)
<i>Matched index return × ln(1 + Bond maturity)</i>	0.4508*** (0.006)	0.2966*** (0.010)	0.5705*** (0.014)	0.1863*** (0.031)
Bond controls				
<i>Mutual fund ownership</i>	0.0071*** (0.003)	−0.0057** (0.002)	0.0032 (0.004)	0.0027 (0.008)
<i>Insurance ownership</i>	0.0058*** (0.001)	−0.0044*** (0.001)	0.0009 (0.002)	0.0101 (0.006)
<i>Turnover</i>	−0.0115*** (0.003)	0.0047 (0.004)	−0.0056 (0.004)	−0.0137 (0.015)
<i>Fraction of zero trading days</i>	0.0002 (0.001)	0.0039*** (0.001)	−0.0003 (0.001)	0.0016 (0.006)
<i>Effective spread</i>	0.1084*** (0.038)	0.0270 (0.049)	0.0375 (0.045)	0.1835 (0.189)
<i>ln(1 + Bond maturity)</i>	−0.0039*** (0.000)	0.0038*** (0.000)	−0.0092*** (0.000)	−0.0001 (0.001)
<i>ln(1 + Bond issue size)</i>	−0.0002 (0.000)	−0.0003 (0.000)	−0.0006 (0.000)	−0.0006 (0.002)
<i>ln(1 + Bond age)</i>	−0.0018*** (0.000)	−0.0001 (0.000)	−0.0005 (0.000)	−0.0008 (0.001)
<i>Investment grade</i>	0.0012 (0.002)			
<i>Upgrade</i>	0.0027*** (0.001)	0.0032*** (0.001)	0.0030*** (0.001)	0.0059 (0.008)
<i>Downgrade</i>	−0.0031 (0.002)	0.0003 (0.001)	−0.0245 (0.021)	0.0021 (0.005)
Issuer × Month fixed effects	YES	YES	YES	YES
Observations	41,516	15,454	21,165	4,897
R-squared	.740	.775	.804	.781

(b) Table 9, panel A

Table 9

(Continued)

B. Differences in the effect of top fund ownership on fed eligible and ineligible corporate bonds

	(1) IG	(2) IG (2–8 years)	(3) IG (4–6 years)
<i>Flow-induced fire sales (FIFS)</i>	−8.5291*** (1.157)	−3.4992** (1.594)	−4.2992* (2.216)
<i>Top fund ownership</i>	−0.0348*** (0.011)	−0.0067 (0.014)	−0.0317 (0.026)
<i>FIFS × Top fund ownership</i>	68.4108*** (21.151)	44.6281* (25.148)	64.7437** (32.095)
<i>Fed eligible</i>	−0.0052*** (0.000)	−0.0017*** (0.001)	−0.0013 (0.001)
<i>FIFS × Fed eligible</i>	10.2739*** (1.323)	4.6358** (1.902)	−0.1303 (2.861)
<i>Top fund ownership × Fed eligible</i>	0.0162 (0.010)	0.0143 (0.013)	0.0422 (0.026)
<i>FIFS × Top fund ownership × Fed eligible</i>	−83.6513*** (22.872)	−46.8364* (28.309)	−54.2217 (39.338)
Market controls	YES	YES	YES
Bond controls	YES	YES	YES
Issuer fixed effects	YES	YES	YES
Month fixed effects	YES	YES	YES
Observations	37,636	15,612	5,249
R-squared	.559	.527	.564

Panel A reports OLS estimates for panel regressions of bond returns on flow-induced fire sales, top fund ownership, and their interaction following the Fed's intervention through the SMCCF. The sample includes 3,049 investment-grade eligible bonds, 4,978 investment-grade ineligible bonds, and 1,430 high-yield bonds. Eligible fallen angels, that is, 164 bonds that were downgraded from investment-grade to below investment-grade after March 2020, are excluded from the sample. The Fed's intervention period spans April 2020 to September 2020. Different subsets of corporate bonds are considered, as indicated on top of each column. All columns include issuer-month fixed effects. Panel B reports OLS estimates for panel regressions of bond returns on flow-induced fire sales, top fund ownership, Fed eligibility, and their interactions during the Fed's intervention period. The samples include both Fed eligible and ineligible investment-grade corporate bonds, but with progressively narrower maturity bands around the 5-year maturity cutoff for Fed eligibility, as indicated on the top of each column. All columns include issuer and month fixed effects. In both panels, for each bond-month observation, the dependent variable, bond return, is calculated as the percentage change in VWAP of the bond from the last day on which there are transactions in the previous month to the last day on which there are transactions in the current month. Only returns calculated from VWAP that lie in the last 10 days of each month are used. Appendix B defines all variables. White's heteroscedasticity robust standard errors are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

(a) Table 9, panel B

Table 10

Bond return exposure to aggregate outflows and ownership concentration

	(1) ALL	(2) IG	(3) HY	(4) HY	(5) HY
<i>Top fund ownership</i>	0.0001 (0.003)	−0.0027 (0.002)	0.0007 (0.003)	−0.0075*** (0.003)	−0.0075*** (0.003)
<i>Aggregate flow D1 (AFD1) × Top fund ownership</i>	0.0146 (0.016)	0.0309*** (0.011)	0.0141 (0.023)	0.0415** (0.017)	0.0416** (0.017)
<i>AFD1 × Mutual fund ownership</i>	−0.0192** (0.009)	−0.0357*** (0.010)	−0.0049 (0.011)	−0.0222*** (0.008)	−0.0221*** (0.008)
<i>AFD1 × JLSW fragility</i>					−0.0009 (0.003)
Market controls	YES	YES	YES	YES	YES
Bond controls	YES	YES	YES	YES	YES
Issuer fixed effects	YES	YES	YES	YES	YES
Month fixed effects	YES	YES	YES	YES	YES
Issuer × Month fixed effects	NO	NO	NO	YES	YES
Observations	911,316	640,885	270,355	189,633	189,633
R-squared	.352	.432	.310	.779	.779

This table reports OLS estimates for panel regressions of bond returns on a large aggregate outflow dummy, top fund ownership, and their interaction. For each bond-month observation, the dependent variable, bond return, is calculated as the percentage change in VWAP of the bond from the last day on which there are transactions in the previous month to the last day on which there are transactions in the current month. Only returns calculated from VWAP in the last 10 days of each month are used. The large aggregate outflow dummy equals one for calendar months in which aggregate flows to all taxable bond mutual funds are in the bottom decile during 1/2003 – 12/2019. Top fund ownership is as of the end of month $t-1$. All variables are defined in Appendix B. All columns include market controls. All columns also include bond controls, all as of month $t-1$, as listed in Table 7 Panel A. Columns (1) – (3) include month and issuer fixed effects. Columns (4) – (5) include issuer-month fixed effects. Standard errors, clustered by month, are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

In column 5, we control for the proxy for fragility of Jiang et al. (2022), which

(b) Table 10

5.6 Tables 11–13: Fund performance, flow fragility, and peer spillovers

Read:

- Table 11 shows that funds with larger LARGEFRAC have higher alpha, especially in periods of aggregate outflows. In the bottom decile of aggregate flows, the coefficient is 0.2316* for all funds and 0.3187** for

high-yield-focused funds.

- Table 12 shows that flows are less sensitive to negative alpha for high-LARGEFRAC funds. The interaction $\text{Alpha} \times \text{LARGEFRAC}$ is -0.2720^* for all funds and -0.3135^* for high-yield-focused funds.
- Table 13 shows that LOWLARGEFRAC peer selling pressure has stronger negative effects on peer performance and flows than HIGHLARGEFRAC peer selling pressure.

Economic message. Stabilizing trading benefits the funds themselves and reduces contagion to overlapping funds. The ownership structure of the mutual fund industry shapes both direct fire-sale losses and spillovers.

Table 11
Fund performance and portfolio fraction of large positions

	Full sample period			Bottom decile of aggregate flows			Bottom quintile of individual flows		
	(1) ALL	(2) IG focused	(3) HY focused	(4) ALL	(5) IG focused	(6) HY focused	(7) ALL	(8) IG focused	(9) HY focused
Top decile positions/TNA (LARGEFRAC)	0.0699*** (0.025)	0.0661* (0.038)	0.1060** (0.050)	0.2316* (0.136)	0.1574 (0.217)	0.3187** (0.150)	0.0347 (0.081)	-0.2046 (0.229)	0.0596 (0.089)
Fund controls	YES	YES	YES	YES	YES	YES	YES	YES	YES
Fund category $\times$ Month fixed effects	YES	YES	YES	YES	YES	YES	YES	YES	YES
Observations	52,652	16,523	36,129	4,945	1,521	3,424	10,781	2,528	8,253
R-squared	.504	.583	.497	.514	.687	.473	.432	.584	.433

This table reports OLS estimates for panel regressions of a fund's alpha (in percent) on its portfolio fraction of positions that are in the top decile of position size (LARGEFRAC). For each fund f in month t , the dependent variable, alpha, is the fund's monthly return minus the benchmark return, estimated from the following equation:

$$R_{f,t} - \text{Risk free rate}_t = \alpha + [\beta_{f,STK} STK_t + \beta_{f,BOND} BOND_t + \beta_{f,DEF} DEF_t + \beta_{f,OPTION} OPTION_t] + \varepsilon_{f,t},$$

where $R_{f,t}$ is the fund's monthly return, STK is the excess return on the CRSP value-weighted stock index, $BOND$ is the excess return on the U.S. aggregate bond index, DEF is the return spread between the high-yield bond index and the intermediate government bond index, and $OPTION$ is the return spread between the GNMA mortgage-backed security index and the intermediate government bond index. All bond indices are from Bank of America Merrill Lynch and are downloaded from DataStream. The parameters, $\beta_{f,STK}$, $\beta_{f,BOND}$, $\beta_{f,DEF}$, and $\beta_{f,OPTION}$ are estimated on a rolling basis. For alpha in month t , the estimation period is from months $t-18$ to $t-1$. All of the independent variables, including LARGEFRAC, are as of the end of month $t-1$. Columns 1–3 are for the full sample period. The subsamples in columns 4–6 include only months in which aggregate flows to all taxable bond mutual funds are in the bottom decile during January 2003 to December 2019. The subsamples in columns 7–9 include only fund-month observations for which individual fund flows are in the bottom quintile, based on pooled sorting across all fund-month observations. Appendix B defines all other variables. All columns include Morningstar's fund category-month fixed effects, and fund controls, including lagged flows, lagged alpha, broker affiliation dummy, portfolio time-varying characteristics (% cash, % government bonds, % corporate bonds, average bond turnover, average fraction of zero trading days, average effective spread, average coupon rate, average credit rating, effective duration, natural logarithm of one plus the average bond issue size, and natural logarithm of one plus the average bond age), and fund time-varying characteristic (age, size, family size, institutional share class fraction, and average maximum rear load). Standard errors, clustered by fund, are in parentheses. $*p < .1$; $**p < .05$; $***p < .01$.

Figure 8: Table 11: fund performance and portfolio fraction of large positions

Table 12
Flow sensitivity to negative performance and portfolio fraction of large positions

	(1) Alpha < 0; ALL	(2) Alpha < 0; IG-focused	(3) Alpha < 0; HY-focused
Alpha (0.01=1%)	0.2935** (0.112)	0.2541** (0.103)	0.3273** (0.146)
LARGEFRAC	0.0019 (0.003)	0.0023 (0.006)	0.0035 (0.003)
Alpha $\times$ LARGEFRAC	-0.2720* (0.145)	-0.1285 (0.272)	-0.3135* (0.170)
Fund controls	YES	YES	YES
Month fixed effects	YES	YES	YES
Observations	27,597	9,055	18,542
R-squared	.269	.264	.303

This table reports OLS estimates for panel regressions of fund flows on fund alpha, LARGEFRAC, and their interaction. The samples include only fund-month observations for which the lagged alphas are negative. For each fund-month observation, the dependent variable, Flow, is computed at month t , while all of the independent variables, including LARGEFRAC, are as of the end of month $t-1$ or prior. Appendix B defines all other variables. All columns include month fixed effects, and fund controls, including lagged flows, lagged alpha, broker affiliation dummy, portfolio time-varying characteristics (% cash, % government bonds, % corporate bonds, average bond turnover, average fraction of zero trading days, average effective spread, average coupon rate, average credit rating, effective duration, natural logarithm of one plus the average bond issue size, and natural logarithm of one plus the average bond age), and fund time-varying characteristic (age, size, family size, institutional share class fraction, and average maximum rear load). Standard errors, clustered by fund, are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

(a) Table 12

Table 13
Spillovers of flow-induced fire sales on performance and flows of peer funds

A. Spillover effects on peers' performance

	(1) ALL	(2) ALL	(3) IG focused	(4) HY focused	(5) ALL
[A] LOWLARGEFRAC peer selling pressure ($\times 100$)	-1.3673*** (0.266)	-1.3967*** (0.240)	0.6682 (0.660)	-1.4488*** (0.367)	-1.1734*** (0.301)
HIGHLARGEFRAC peer selling pressure ($\times 100$)	-0.2689 (0.193)	-0.5284** (0.230)	-0.4054 (0.462)	-0.1462 (0.323)	
HIGHLARGEFRAC peer selling pressure from "large" pos. ($\times 100$)					0.0684 (0.445)
[B] HIGHLARGEFRAC peer selling pressure from "small" pos. ($\times 100$)					-2.5720 (1.997)
(Time-varying) fund controls	NO	YES	YES	YES	YES
Fund fixed effects	YES	YES	YES	YES	YES
Month fixed effects	YES	YES	YES	YES	YES
F-statistic for test of whether [A] = [B]					0.92
Observations	54,313	53,209	16,771	36,438	53,209
R-squared	.356	.427	.555	.440	.428

B. Spillover effects on peers' flows

[A] LOWLARGEFRAC peer selling pressure ($\times 100$)	-0.1876*** (0.016)	-0.1639*** (0.014)	-0.1064*** (0.033)	-0.1081*** (0.020)	-0.1573*** (0.014)
HIGHLARGEFRAC peer selling pressure ($\times 100$)	-0.0575*** (0.008)	-0.0578*** (0.007)	-0.0495*** (0.019)	-0.0419*** (0.011)	
HIGHLARGEFRAC peer selling pressure from "large" pos. ($\times 100$)					-0.0404*** (0.010)
[B] HIGHLARGEFRAC peer selling pressure from "small" pos. ($\times 100$)					-0.1763*** (0.059)
(Time-varying) fund controls	NO	YES	YES	YES	YES
Fund fixed effects	YES	YES	YES	YES	YES
Month fixed effects	YES	YES	YES	YES	YES
F-statistic for test of whether [A] = [B]					0.65
Observations	63,659	53,220	16,774	36,446	53,220
R-squared	.209	.322	.326	.340	.322

This table reports OLS estimates for panel regressions of fund alpha (in percent) (panel A) and fund flows (panel B) on peer selling pressures. In columns 1–4 of both panels, the peer selling pressures are separated into those from peers with LARGEFRAC in and not in the top decile in each month, as denoted by HIGHLARGEFRAC and LOWLARGEFRAC. In column 5, the selling pressures from peers in the HIGHLARGEFRAC group are further split into those from positions in and not in the top decile of position size, as denoted by "large" and "small" positions. For each fund-month observation, the dependent variables, Alpha and Flow, are computed at month t , as described in Tables 11 and 12. Peer selling pressure is calculated as the position-weighted average of FIFs from a given type of positions from a given type of peers in a given bond, where the position weight is as of month $t-1$ but the peer flows that drive the selling pressure are as of month t . All other independent variables are as of the end of month $t-1$ or prior, and are defined in Appendix B. Following Falato et al. (2021), all columns include fund and month fixed effects. Columns 2–5 also include fund controls, including lagged flows, lagged alpha, portfolio time-varying characteristics (% cash, % government bonds, % corporate bonds, average bond turnover, average fraction of zero trading days, average effective spread, average coupon rate, average credit rating, effective duration, natural logarithm of one plus the average bond issue size, and natural logarithm of

(b) Table 13

6 Short summary

- The paper studies fire-sale behavior by corporate bond mutual funds.
- Funds with large shares of bond issues internalize their own price impact.
- The theory predicts that these funds sell large issue-share positions less during redemptions.
- The main trading regression confirms this prediction using fund-by-period and bond-by-period fixed effects.
- Mechanism tests show stronger effects when funds have fewer liquid assets, when bonds are less liquid, and when same-family funds also hold large positions.
- Fund merger IV results support a causal interpretation.
- The Fed's SMCCF test shows that stabilizing trading weakens for Fed-eligible bonds, consistent with reduced expected spillovers.
- Bond return regressions show that ownership concentration lowers exposure to flow-induced fire sales.
- Funds with more large positions perform better in turmoil and have less fragile flows.
- Selling pressure from high-LARGEFRAC funds creates smaller peer spillovers than selling pressure from low-LARGEFRAC funds.

7 Conclusion

7.1 Core conclusion

Giannetti and Jotikasthira show that concentrated ownership in corporate bond mutual funds can reduce fire-sale fragility. Funds that own large shares of outstanding bond issues sell those bonds less when facing redemptions, because they internalize the price impact of their trades.

7.2 Economic intuition

A large owner is not atomistic. Selling a large position depresses the bond price and immediately reduces the value of the fund's remaining holdings. This gives the fund a private incentive to avoid selling the bond aggressively. The result is a stabilizing force: concentrated ownership can dampen price pressure and reduce contagion.

7.3 Takeaway

The paper changes the usual interpretation of large mutual funds. Large funds may create risk when they suffer idiosyncratic redemptions, but when many funds face correlated outflows, large positions can also stabilize prices by making fund managers internalize fire-sale externalities. Regulators should therefore pay attention not only to large funds but also to the systemic consequences of many small funds with correlated flows.